Challenges in real-time embedded IoT Command Recognition

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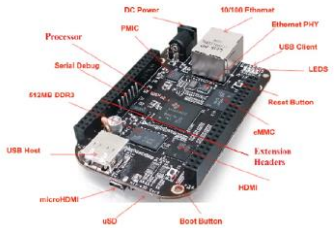
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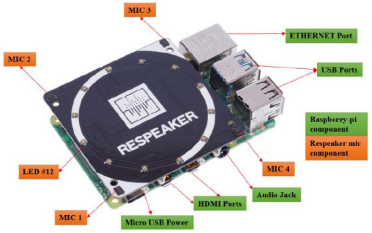
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Presentation Outline

- Automatic Speech Recognition: Challenges
- Current Commercial ASR Systems: Embedded
- Computational Platforms:
  - Beaglebone Black & Raspberry Pi 3



- Data Portfolio: IoT “Command Recognition”
- Algorithm Development:
  - CNN
  - DSCNN



- Real-time system Deployment: Field Opeatrions
- Evaluation Results
- Summary & References

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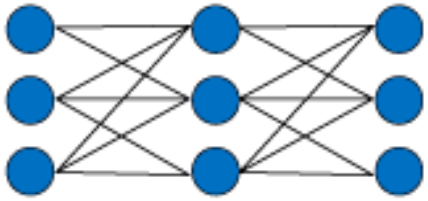
Automatic Speech Recognition



Speech Signal

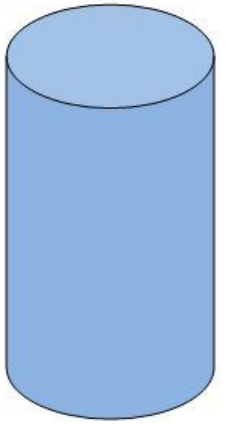


Feature Extraction



Model training

Training



Trained Model



Unknown Utterance



Testing

IoT COMMAND RECOGNITION

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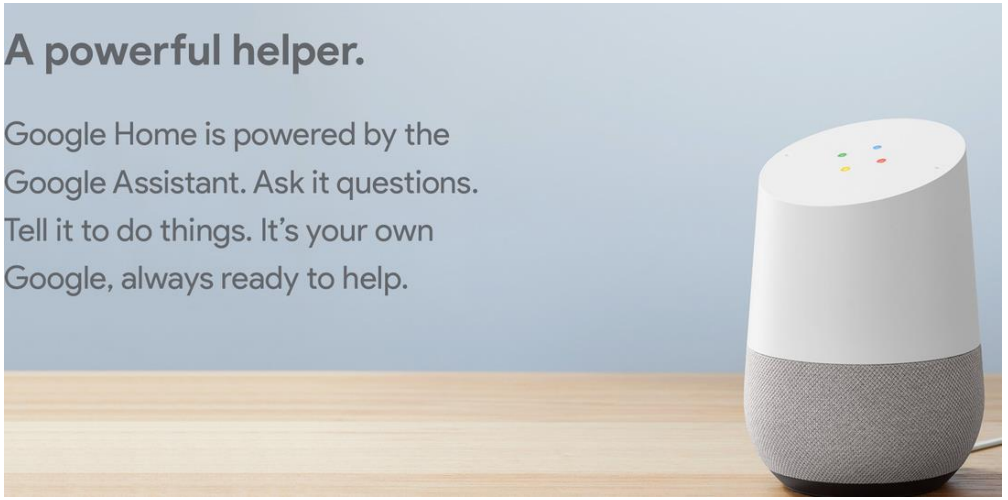
Commercial ASR systems

Echo Dot



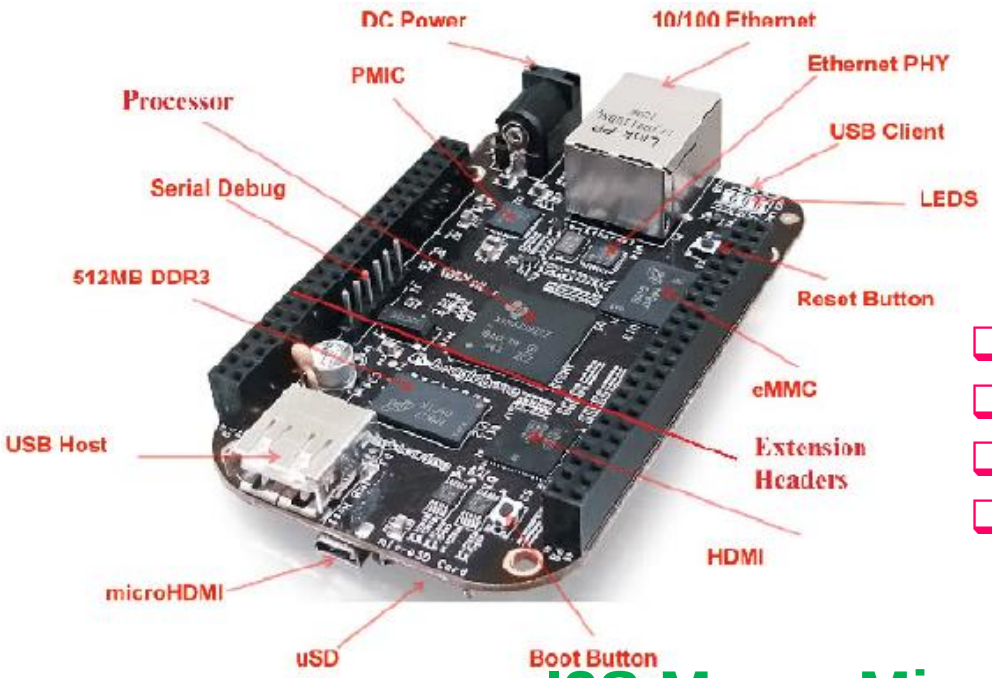
- Supports commands such as music “play music”, “set timer” ,“translate phrase”
- Recognizes different speakers and personalize response
- Wi-Fi supporting Integration with other IoT devices in smart home environment

Google Home



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Computational Platforms



Beaglebone Black

TEXAS INSTRUMENTS

- CPU : Arm Cortex-A8, 32-bit
- Memory: 512MB DDR3 RAM
- Storage: 4GB flash storage
- CPU Speed: 1GHz

I2S Mems Microphone



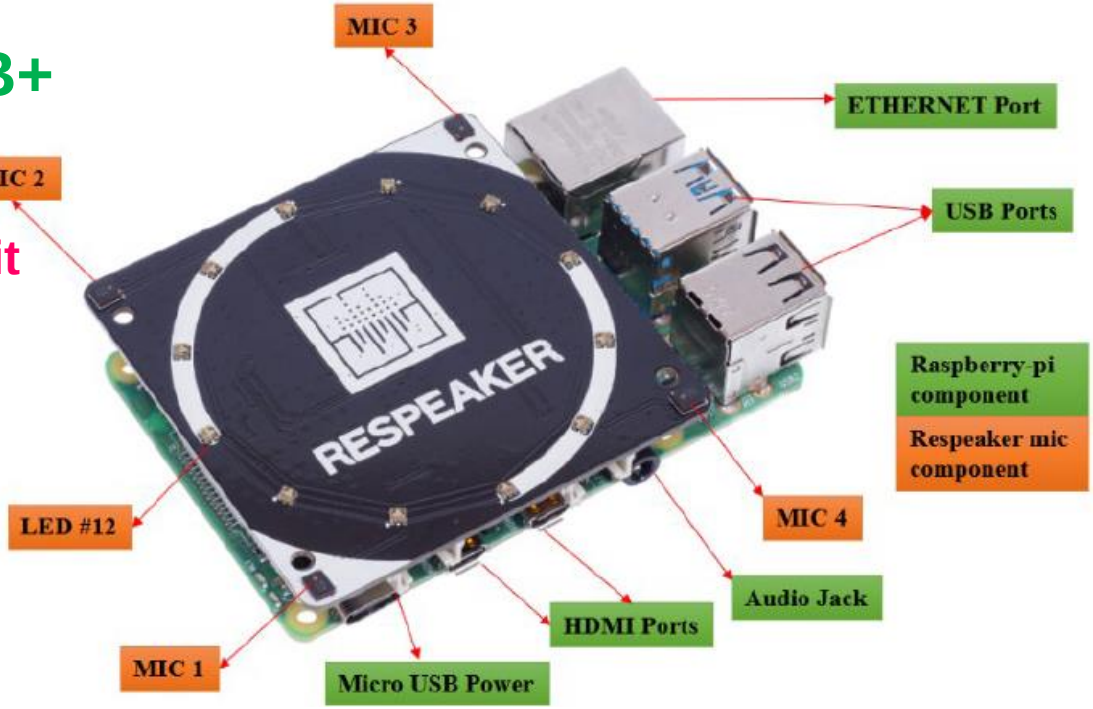
- MIC: 2 Digital Output Microphones
- Omnidirectional
- Dimensions: 3.50 x 2.65 x 0.98 mm
- Requires additional circuitry to integrate with microcontrollers

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Computational Platforms

Raspberry Pi 3 Model B+

- CPU : Arm Cortex-A53, 64-bit
- Memory: 1GB RAM
- Storage: 4GB flash storage
- CPU Speed: 1.2GHz



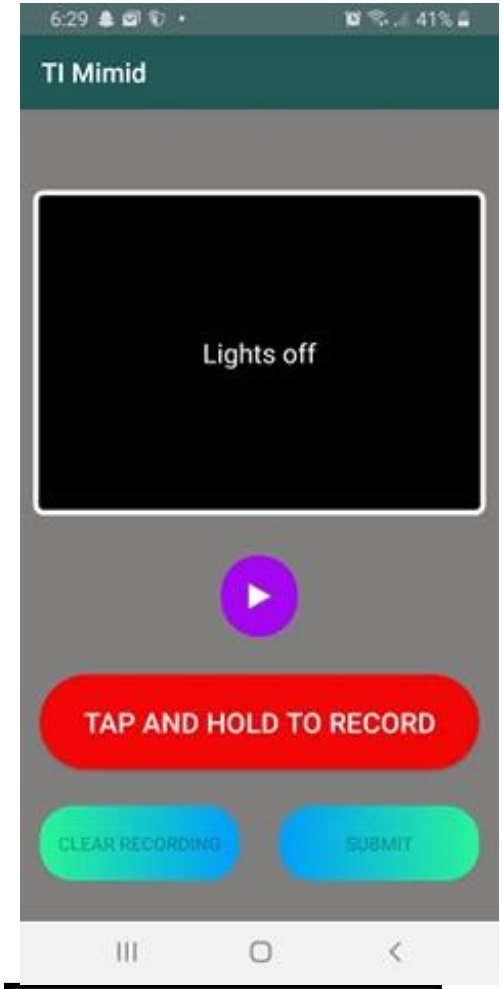
ReSpeaker 4-Mic array

- MIC: 4 Analog microphones
- Omnidirectional
- Dimensions: 65mm x65mm x9mm
- 12 APA102 programable RGB LEDs

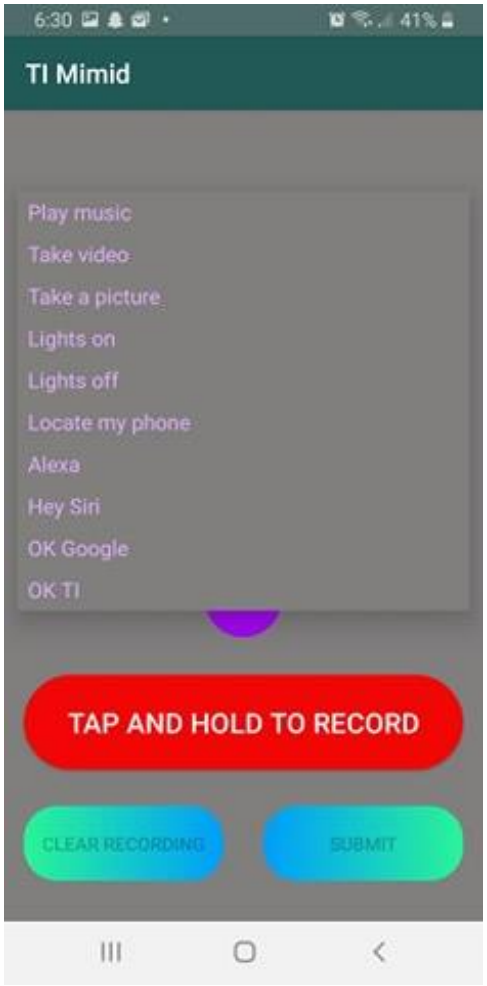
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Data Portfolio

"TI Mimid" Data Collection App



- Prompts user to record commands
- Flexibility allowing user to re-record before submission
- Recordings stored in an online database
- Each user allocated a unique ID within the database to protect user anonymity



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Data Portfolio

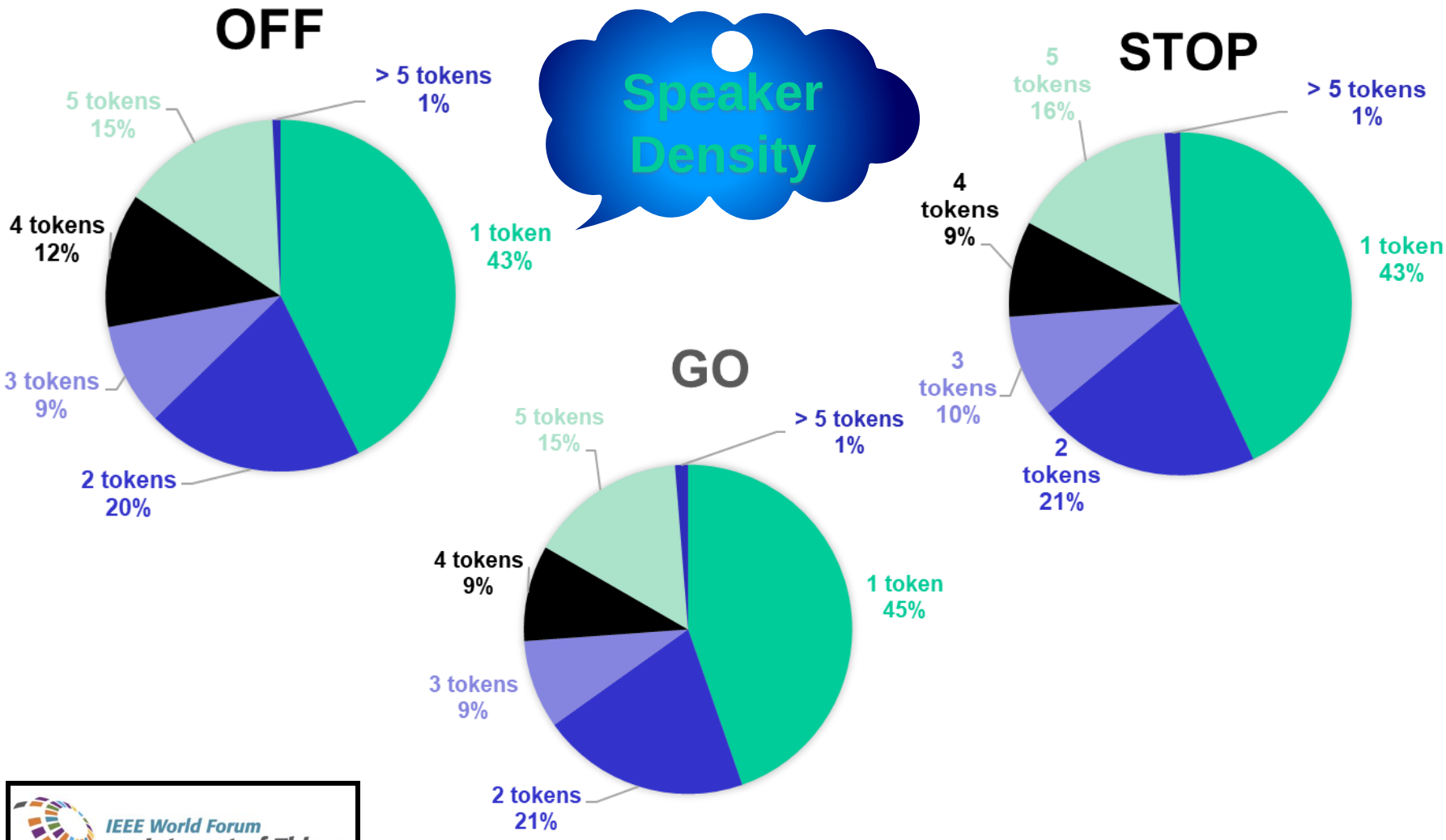


Google AI Blog

- Google Speech Commands Dataset
- Crowdsourced collection of 65,000 1-sec speech signals
- Metadata unavailable so a data portfolio was developed based on Speaker Token Density allowing the visualization of each speaker's contribution to the dataset as well as the effect on the robustness of the system

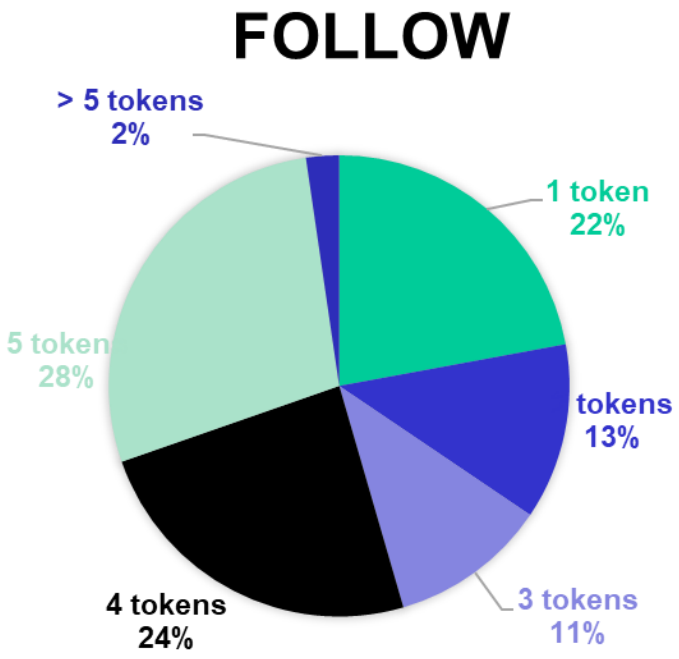
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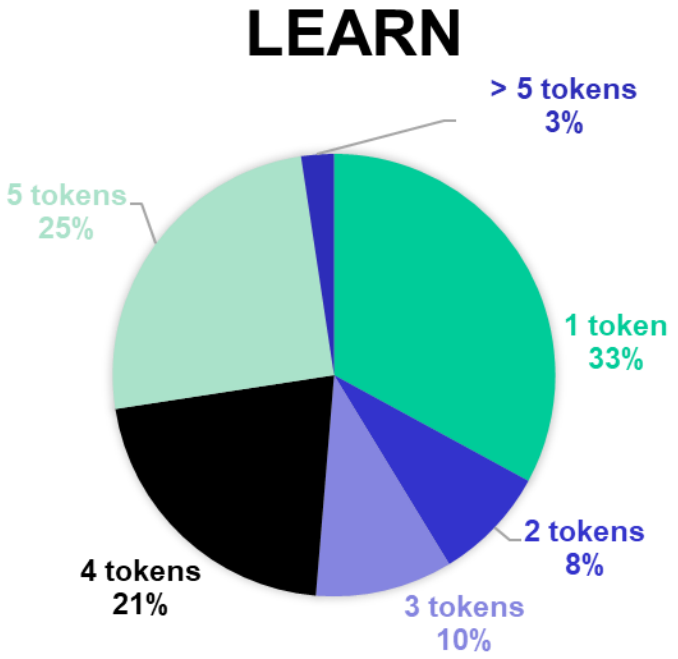
Data Portfolio (Commands & Speaker Tokens)



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Data Portfolio (Commands & Speaker Tokens)



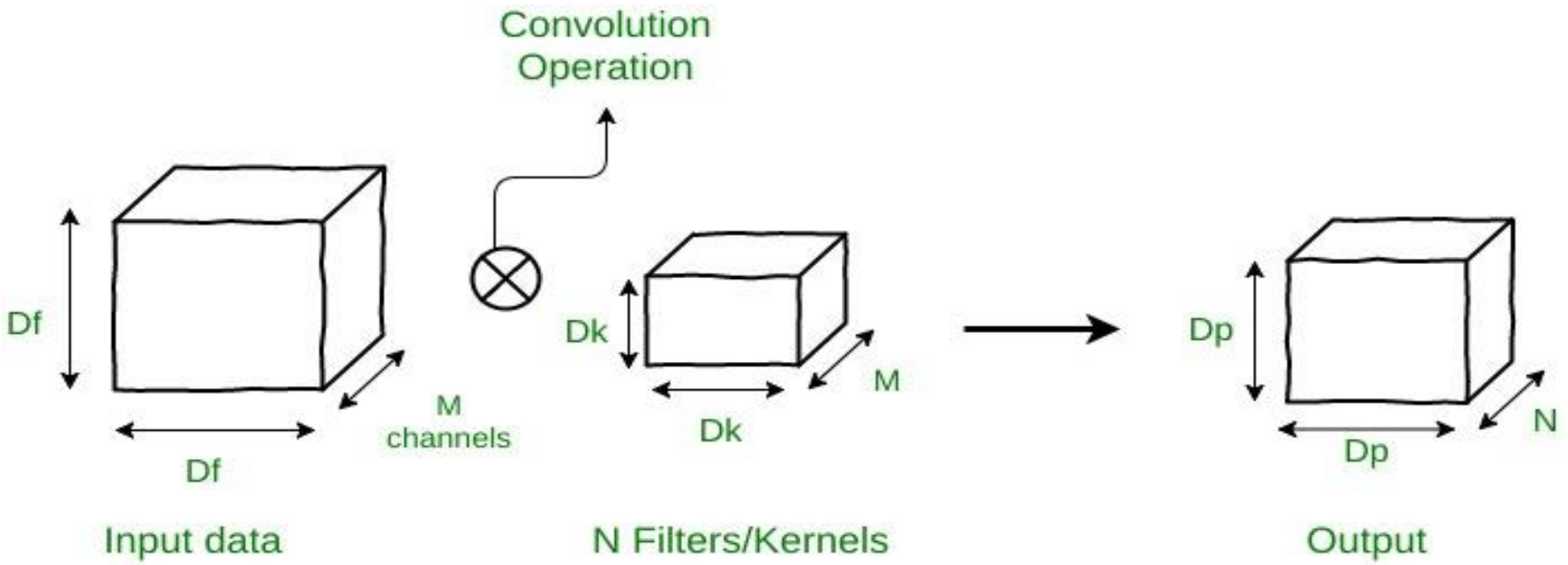


- 1300 pseudo random tokens selected for training each model
- Close to half the speakers provided only 1 utterance
- Diversity in speakers supports ASR whereas diversity in “same-speaker” tokens more prominent in SID domain

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Algorithm Development

Convolutional Neural Network (CNN)

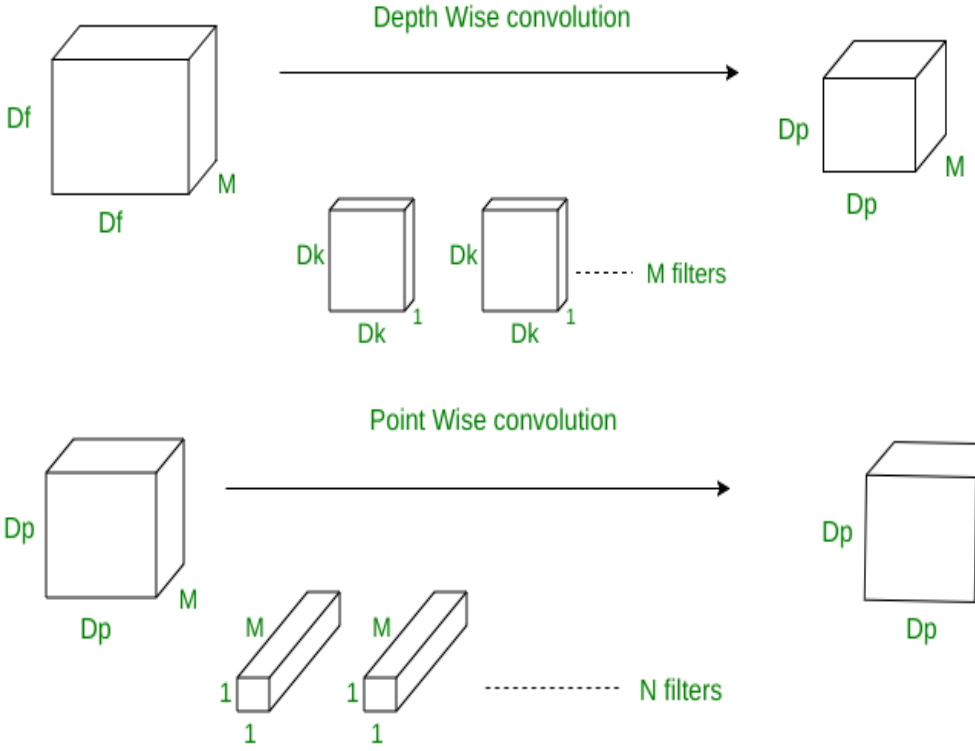


- The number of multiplications in 1 convolution operation i.e. size of filter = *Dk* x *Dk* x *M*
- Filter slides vertically and horizontally "*Dp*" times
- Resulting # of multiplications = $N \times Dp^2 \times Dk^2 \times M$

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Algorithm Development

Depth-Wise Separable Convolutional Neural Net (DSCNN)



- The number of multiplies in 1 conv. operation
  $= Dk \times Dk$
- Resulting # of multiplications
  $= Dp^2 \times Dk^2 \times M$

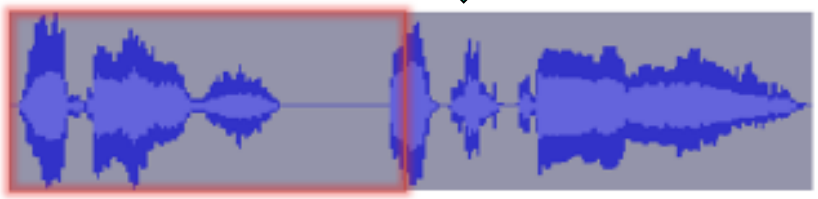
- The number of multiplies in 1 conv. operation = $1 \times M$
- Resulting # of multiplications
  $= Dp^2 \times N \times M$

- Total multiplications = Depth-wise conv. Multiplies + Point-wise conv. Multiplies
- Total Cost of DSCNN operations = $M \times Dp2 \times (Dk2 + N)$

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System Deployment

Deployed Pipeline



Continuous Raw Audio

1 sec buffer length

MFCC Extraction → Inference engine → P≥0.5 → LED Activation / Do Nothing



Deployed System w/ Integrated LED functionality

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Experimental Results: IoT Command Recognition

Model / Command	Training Algorithm	Evaluated Accuracy
STOP	CNN	94.63%
	DSCNN	98.34%
LEARN	CNN	96.13%
	DSCNN	94.43%
GO	CNN	92.89%
	DSCNN	98.80%
OFF	CNN	97.30%
	DSCNN	97.87%
FOLOW	CNN	93.75%
	DSCNN	93.67%

AVG: CNN 94.94% DSCNN 97.22%

Summary

- Presented a comparison between two Machine Learning approaches to train an ASR System on a low-level compute platform
- Explored several computational platforms for deployment
- Discussed Dataset Collection app and similarly crowdsourced “Google Command Dataset”
- Each command deployed individually to compare the two ML architectures
- System Deployment pipeline demonstrated along with study results
- Results do not favor a particular Machine learning architecture for commands deployed!

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